LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to directional divergence, spiral divergence, Dutch roll and controllability
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to directional divergence, spiral divergence, Dutch roll and controllability.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagrams and stability charts; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define directional stability in simple terms.
(b) What is the primary function of the rudder?
(c) How does the AFCS assist the pilot in maintaining heading?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Directional Divergence • Causes of Directional Instability • Role of Vertical Stabilizer • AFCS Heading Hold Function • Recovery from Divergence	L & Ds	PPT, IB	15 min
(b)	• Definition of Spiral Divergence • Relationship between Roll and Yaw • Spiral Stability Criteria • AFCS Roll/Yaw Coordination • Pilot Recognition of Spiral Mode	L & Ds	PPT, PDF	20 min
(c)	• Definition of Dutch Roll • Coupled Yaw and Roll Oscillation • Damping Mechanisms • AFCS Yaw Damper Function • Control Effectiveness in Oscillation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS yaw damper mitigates Dutch roll.
(b) Differentiate between directional divergence and spiral divergence.
(c) Describe the pilot's role in managing controllability during oscillations.

12. Home Assignment and Follow Up	Prepare a brief note on the fundamental elements of the flight control system as discussed in LP 15, focusing on how they interact with AFCS to ensure controllability.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need and role of flight control system, fundamental elements of flight control system.	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min